Shivani Kalamadi

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Education

University of California, Davis — Davis, CA
MS and PhD in Computer Science; ***GPA: 3.88/4.00*** — *09/2022 – 06/2027*

SDM College of Engineering and Technology — Karnataka, India
B.E. in Computer Science; ***GPA: 9.00/10.00*** — *08/2015 – 06/2019*

Research

Networks - Internet Measurement — UC Davis
Advisor: Alexander Gamero-Garrido – *ImplyLab* — *06/2024 – Present*

Digital Divide & Internet Access

- Investigated U.S. internet access disparities through a socio-economic lens across 20 metropolitan regions, using five years (2021–2025) of **Ookla** crowd-sourced speed tests.
- Applied **statistical modeling** and **machine learning**: Identified and corrected for sampling bias, and used a Random Forest regression model to correlate socio-economic features with network metrics, validating results with permutation and null-model importance.
- Incorporated the Broadband-Plan Querying Tool (BQT) for subscription plans and validated results using infrastructural and funding differences.
- Extending the framework for India/Brazil for cross-country generalizability, adapting the pipeline to varying administrative geographies and sparser crowd-sourced coverage.

Web Infrastructure & Privacy

- Studying the **hosting sovereignty** and **privacy** of U.S. state and federal web infrastructure: who can observe data in transit, where it is routed and stored, which jurisdiction governs it, and which networks operate the service.
- Developing a pipeline that traces government websites from **IP** to operating **organization**, producing a state-wise map of hosting strategies to identify points of failure across jurisdictions.
- Measuring **first- and third-party trackers** and **resource dependencies** at each level of page load, revealing the extent of external reliance across government web infrastructure.

Application-Level Performance

- Investigating how application-layer performance for web browsing and video streaming tracks **Ookla** measurements, identifying to what extent speed tests are reliable predictors of user experience.
- Leading collection and analysis of YouTube and YouTube Shorts **QoE**, building an automated framework that collects and correlates 20 streaming metrics with speed-test results under controlled network conditions to bridge network-level measurement and real-world application performance.

Reverse-geolocating Community Anchor Institutions

- Led validation of **geographic representativeness** for **Reverse IP Geolocation (RG)**, a framework that infers IP addresses of libraries, schools, and hospitals from public street addresses, checking candidates against DNS records, WHOIS data, broadband provider data, and active measurements.

Publications & Presentations

Stratifying the Digital Divide: Analysis of Socio-Economic Influences on Internet Performance

- Presented paper at **IMC 2025 Student Workshop** - Madison, WI
- Poster accepted at **TMA 2026 PhD School** - Delft, Netherlands
- Poster presented at **SIGCOMM 2026 N2Women** - Denver, CO
- Pre-print: arXiv:2605.30809

Same Speed, Different Experience? Testing the Limits of an Ookla Test

- Poster at **SIGCOMM 2026 N2Women** - Denver, CO

Where's Waldo Library? Using Reverse IP Geolocation to Identify Library IPs

- Pre-print: arXiv:2605.30791

Experience

Oracle Bangalore, India

Senior Cloud Analyst and Consultant – Integration & ERP Systems *08/2019 – 05/2022*

- Worked on Oracle Fusion Applications to manage core business functions such as finance, procurement, and project management, developing a deep understanding of how data is stored and maintained across a large enterprise system.
- Built data pipelines using **SQL** on BI Publisher and Integration Cloud that moved business records between client systems and Oracle Cloud.
- Transformed and aggregated business records - merging, filtering, and reshaping data to client specifications across reporting pipelines.
- Translated client requirements into technical specifications end to end, and led the campus-to-corporate program, mentoring the 2020 and 2021 new-hire cohorts.

Projects

A Study on Browser Extension Ecosystem | *GitHub*

- Compared the permission and privacy models of the top Chrome and Firefox extensions to surface how each browser exposes users to over-permissioned or data-leaking extensions.
- Ran **static analysis** of `manifest.json` permissions and domains via Python and the ExtAnalysis core library, plus **dynamic analysis** of per-extension network traffic.

Abstract Art Using Speech Emotions | *GitHub*

- Explored whether emotion detected from speech could drive generative visual art, mapping recognized emotional states to abstract image generation.
- Paired VQGAN and CLIP as generator and perceptor, and compared CNN, LSTM, and NLP-based speech-emotion recognition models on classification performance.

Custom Operating System and Game | *GitHub*

- Built this OS from scratch on a RISC-V-based game console and developed a snake game to work with the firmware through API support such as I/O Controller APIs, Timer-based APIs, Thread Management, Graphics APIs, and Memory Management APIs.

Skills

Languages: Python, C, Java, JavaScript, SQL

Tools: Git, Docker, AWS, REST & GraphQL APIs, Tableau, BI Publisher, GeoPandas

ML/Data Science: pandas, NumPy, SciPy, scikit-learn

Measurement Data: Ookla, M-Lab, Census, CAIDA

Measurement Tools: Selenium, Chrome DevTools, RIPE Atlas, Team Cymru, IPInfo